Chapter 8 — Physical Optics and Gravitational Waves

Full textbook exercise · PECTAA Physics 11 · No ALP filtering

Q.1 — Multiple Choice Questions (with correct option and reason)

8.1. The phenomenon of polarization of light is:

(a) scattering of light
(b) the property of light to vibrate in a specific plane
(c) ability to travel in a straight line
(d) changing colour

Answer: (b) the property of light to vibrate in a specific plane — Polarization restricts vibrations to one plane.

8.2. Malus's law states that intensity is directly proportional to:

(a) the square of cosine of the angle
(b) the square of sine of the angle
(c) the angle
(d) the inverse of the angle

Answer: (a) the square of cosine of the angle — $I = I_0 \cos^2\theta$.

8.3. The intensity of light when it passes through a polarizer:

(a) increases
(b) decreases
(c) remains the same
(d) becomes zero

Answer: (b) decreases — Unpolarized light loses about half its intensity.

8.4. The angle between the light wave and the analyzer is called:

(a) polarization angle
(b) refraction angle
(c) reflection angle
(d) azimuth angle

Answer: (d) azimuth angle — It is the angle between transmission axes.

8.5. The key purpose of an analyzer in a polarization experiment is:

(a) to polarize the light
(b) to measure the intensity of light
(c) to change direction of light
(d) to filter unwanted light

Answer: (b) to measure the intensity of light — It examines the polarized light and its intensity.

8.6. The mathematical representation of Malus's law is:

(a) $I = I_0 \cos^2\theta$
(b) $I = I_0 \sin^2\theta$
(c) $I = I_0 \tan^2\theta$
(d) $I = I_0 \cot^2\theta$

Answer: (a) $I = I_0 \cos^2\theta$ — Intensity follows amplitude squared.

8.7. Increasing the angle between light wave and analyzer makes intensity:

(a) increase
(b) decrease
(c) remain same
(d) become zero always

Answer: (b) decrease — $\cos^2\theta$ decreases from 0° to 90°.

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8.8. Maximum intensity in a polarization experiment occurs when light wave and analyzer are:

(a) perpendicular **(b)** parallel
(c) at 45° **(d)** at 60°

Answer: (b) parallel — $\cos^2 0° = 1$.

8.9. Unwanted light that interferes with vision is termed:

(a) haze **(b)** glare
(c) contrast **(d)** flare

Answer: (b) glare — Glare is unwanted reflected light.

8.10. Who predicted the existence of gravitational waves?

(a) Galileo Galilei **(b)** Albert Einstein
(c) Isaac Newton **(d)** Leonardo da Vinci

Answer: (b) Albert Einstein — Einstein predicted them in 1916.

8.11. What are gravitational waves?

(a) electromagnetic waves **(b)** mechanical waves
(c) ocean waves **(d)** ripples in the fabric of space-time

Answer: (d) ripples in the fabric of space-time — They are space-time distortions.

8.12. The primary method used to detect gravitational waves is:

(a) optical telescopes **(b)** radio telescopes
(c) LASER interferometry **(d)** gravitational lensing

Answer: (c) LASER interferometry — LIGO uses laser interferometry.

8.13. Which is a primary source of gravitational waves?

(a) binary black hole merger **(b)** solar flares
(c) earthquake **(d)** solar wind

Answer: (a) binary black hole merger — Accelerating compact masses radiate GWs.

Q.2 — Short Answer Questions

8.1. Why are polaroid sunglasses better than ordinary sunglasses?

They block polarized glare reflected from horizontal surfaces such as water and roads, improving contrast and reducing eye strain, not just overall brightness.

8.2. Is light from the sky partially polarized? How?

Yes. Sunlight scattered by air molecules is partially polarized, especially at 90° to the Sun, because scattering favours vibrations perpendicular to the scattering plane.

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8.3. How is Malus's law used in everyday life?

It is used in polarized sunglasses, camera filters, LCD screens and stress analysis, where rotating a polarizer controls transmitted intensity.

8.4. What are the applications of Brewster's angle?

It is used to produce polarized light by reflection, in polarizing windows, laser cavities and glare-reducing optical designs.

8.5. What is space-time curvature?

Mass and energy bend the four-dimensional space-time geometry. Objects follow geodesics in this curved geometry, which we observe as gravity.

Constructed Response Questions

8.1 Write some applications of plane-polarized light.

Polaroid sunglasses, sky photography, LCD displays, stress analysis of transparent models, and controlling light intensity in optical instruments.

8.2 Can a polarizer be used as an analyzer? Give two examples.

Yes. The same polaroid can examine polarized light. Examples: testing sunglasses by rotating them against a reflective glare, and checking stress patterns in a transparent model between crossed polaroids.

8.3 How is Malus's law used in polarized sunglasses? How do they reduce glare?

Reflected glare is partially polarized horizontally. The sunglasses' vertical transmission axis blocks much of it. Rotating the lens changes intensity according to $I = I_0 \cos^2\theta$.

8.4 How would the sky appear if there were no atmosphere?

It would appear dark even in daytime because there would be no air molecules to scatter sunlight.

8.5 What is the significance of detecting gravitational waves?

They open a new way to observe violent cosmic events such as black-hole mergers, test general relativity and study objects that emit little or no light.

8.6 How are tidal forces formed?

They arise because a gravitational field is stronger on one side of a body than the other, producing stretching and compressing. Gravitational waves produce similar oscillating tidal distortions.

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Comprehensive / Long Questions

8.1 Define polarization. How does polarization of electromagnetic waves occur? Classify polarization.

Polarization restricts the electric field vibrations to a particular pattern. It occurs when unpolarized light passes through a polarizer or reflects at Brewster's angle. Types include plane, circular and elliptical polarization.

8.2 How can plane-polarized light be produced and detected? What does it prove?

It is produced by a polaroid or reflection at Brewster's angle and detected by an analyzer. Crossed polaroids block light, proving light is transverse.

8.3 How can polarized light be obtained by reflection? Explain.

At Brewster's angle, $\tan \theta_p = n$, the reflected light is plane-polarized perpendicular to the plane of incidence. The reflected and refracted rays are 90° apart.

8.4 State Malus's law. Explain the intensity formula.

Malus's law: $I = I_0 \cos^2\theta$. The analyzer passes amplitude $A_0 \cos \theta$; since intensity is proportional to amplitude squared, the transmitted intensity is $I_0 \cos^2\theta$.

8.5 What is a polaroid? Explain two main applications of polarization.

A polaroid is a sheet that transmits only one vibration direction. Applications: sunglasses reduce glare, and stress analysis shows coloured fringes in stressed transparent models.

8.6 What are gravitational waves? Describe their basic types.

Gravitational waves are ripples in space-time travelling at c. Types: continuous from spinning masses, compact binary inspiral, burst from violent events, and stochastic background.

8.7 What is an interferometer? Describe the basic LIGO interferometer.

An interferometer splits a laser beam into perpendicular arms, reflects it from mirrors and recombines it. A gravitational wave changes arm lengths differently, disturbing destructive interference and producing a signal. LIGO has 4 km arms.

8.8 What is optical activity? Discuss it.

Optically active substances such as quartz and sugar solution rotate the plane of polarization. The rotation depends on thickness, concentration and wavelength, and is used in polarimetry.

Numerical Problems (fully worked)

8.1 Unpolarized light of intensity I_0 is incident on a polarizing sheet. Find intensity not transmitted.

Given: incident intensity = I_0

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Formula: transmitted = $I_0/2$

Not transmitted = $I_0 - I_0/2 = I_0/2$.

Answer: $I_0/2$

8.2 Polarized light 100 W m⁻² passes through a polarizer at 45°. Find transmitted intensity.

Given: $I_0 = 100\ W\ m^{-2}$, $\theta = 45°$

Formula: $I = I_0 \cos^2\theta$

$I = 100 \times 0.5 = 50\ W\ m^{-2}$.

Answer: 50 W m⁻²

8.3 A light wave amplitude 10 units passes a polarizer at 30°. Find transmitted amplitude.

Given: $A_0 = 10$, $\theta = 30°$

Formula: $A = A_0 \cos\theta$

$A = 10 \times 0.866 \approx 8.66$ units.

Answer: 8.66 units

8.4 What angle reduces intensity by 85%?

Given: transmitted fraction = 0.15

Formula: $\cos^2\theta = 0.15$

$\theta = \cos^{-1}(\sqrt{0.15}) \approx 67.5°$.

Answer: 67.5°

8.5 Unpolarized light 15 W m⁻² passes two polarizers at 50° and 20° from vertical. Find intensity after each.

Given: $I_0 = 15\ W\ m^{-2}$, angle between axes = 30°

Formula: first: $I_0/2$; second: $I_1 \cos^2 30°$

After first: 7.5 W m⁻².

After second: $7.5 \times 0.75 \approx 5.6\ W\ m^{-2}$.

Answer: 7.5 W m⁻², 5.6 W m⁻²

8.6 Two polarizing sheets are parallel for maximum intensity. Through what angle must one be rotated to halve intensity?

Given: $I/I_0 = 0.5$

Formula: $\cos^2\theta = 0.5$

$\theta = 45°$.

Answer: 45°

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8.7 Glass n = 1.5 in air. Find polarizing angle and angle of refraction.

Given: n = 1.5

Formula: $\tan \theta_p = n$, $\theta_p + \theta_r = 90°$

$\theta_p = \tan^{-1}(1.5) \approx 56.3°$.

$\theta_r = 90° - 56.3° = 33.7°$.

Answer: 56.3°, 33.7°

8.8 At what angle of incidence is light reflected from water completely polarized? n ≈ 1.33.

Given: n = 1.33

Formula: $\tan \theta_p = n$

$\theta_p = \tan^{-1}(1.33) \approx 53°$.

Answer: 53°

8.9 Unpolarized light passes four polarizers, each rotated 30° from the previous. Find percentage transmitted.

Given: first transmits half; each next $\cos^2 30° = 0.75$

Formula: fraction = $\frac{1}{2} \times (0.75)^3$

Fraction = $0.5 \times 0.421875 \approx 0.211 = 21\%$.

Answer: 21%

8.10 Polarizer and analyzer axes at 60°. Find fraction of initial intensity emerging.

Given: $\theta = 60°$

Formula: $I/I_0 = \cos^2\theta$

$\cos^2 60° = 0.25$.

Answer: 0.25

8.11 Gravitational waves wavelength 3000 km. Find frequency.

Given: $\lambda = 3\times10^6$ m, $c = 3\times10^8$ m s^{-1}

Formula: $f = c/\lambda$

$f = 3\times10^8 / 3\times10^6 = 100$ Hz.

Answer: 100 Hz

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